Experiment 1

Student Name: Yash Kotra
UID: 23BCS10599
Branch: BE-CSE
Section: Group: KRG-3A
Semester: 6
Date of performance: 08-01-2026
Subject name: System Design
Subject code: 23CSH-314

AIM:

To design, implement, and evaluate a simple URL redirection system that converts long URLs into short URLs, and to measure system performance (latency and throughput) under varying request loads.

OBJECTIVE:

1. To understand the working of a URL shortening service.
2. To design REST APIs for URL creation and redirection.
3. To identify suitable database and server choices.
4. To analyze different short URL generation techniques.
5. To understand scalability challenges and their solutions.

PROBLEM STATEMENT:

Design a URL Shortener system that converts long URLs into short URLs and redirects users from short URLs to original long URLs with low latency, high availability, and scalability.

1.Requirements:-

A. Functional Requirements:-

1. Generate a short URL from a long URL.
2. Support custom URLs (optional).
3. Support URL expiration (default and custom).
4. Redirect short URL to the original long URL. 5. User registration and login using REST APIs.

B. Non-Functional Requirements:-

1. Low Latency: URL creation and redirection < 20ms
2. Scalability:
   - 100M daily active users
   - 1B shortened URLs
3. Availability: System should be available 24×7.
4. Uniqueness: Each short URL must be unique.
5. CAP Theorem:
   - Availability > Consistency

     Eventual Consistency model.

System Entities

1. User 2. Long URL 3. Short URL

2.API Design:-

1. Create Short URL Endpoint: POST /v1/url

Request Body:

```
{
  "longURL": "string",
  "customURL": "string (optional)",
  "expirationDate": "date"
}
```

Response:

```
{
  "shortURL": "https://short.ly/2bi"
```

}

2. Redirect Short URL

Endpoint: GET /v1/url/{shortURL}

Action: Redirects to original long URL.

3. User APIs

- POST /v1/register
- POST /v1/login

3.Database Design:-

Table T1 – User Metadata

Column	Description
user_id	Primary Key
name	User Name
email	Unique
password	Encrypted
created_at	Timestamp

Table T2 – URL Table

Column	Description
id	Primary Key
shortURL	Unique
longURL	Original URL
customURL	Optional
expirationDate	Expiry Time
user_id	Foreign Key

4.High Level Design (HLD) Components:-

1. Client (Browser / App)
2. Server (Business Logic)
3. Database (Storage)

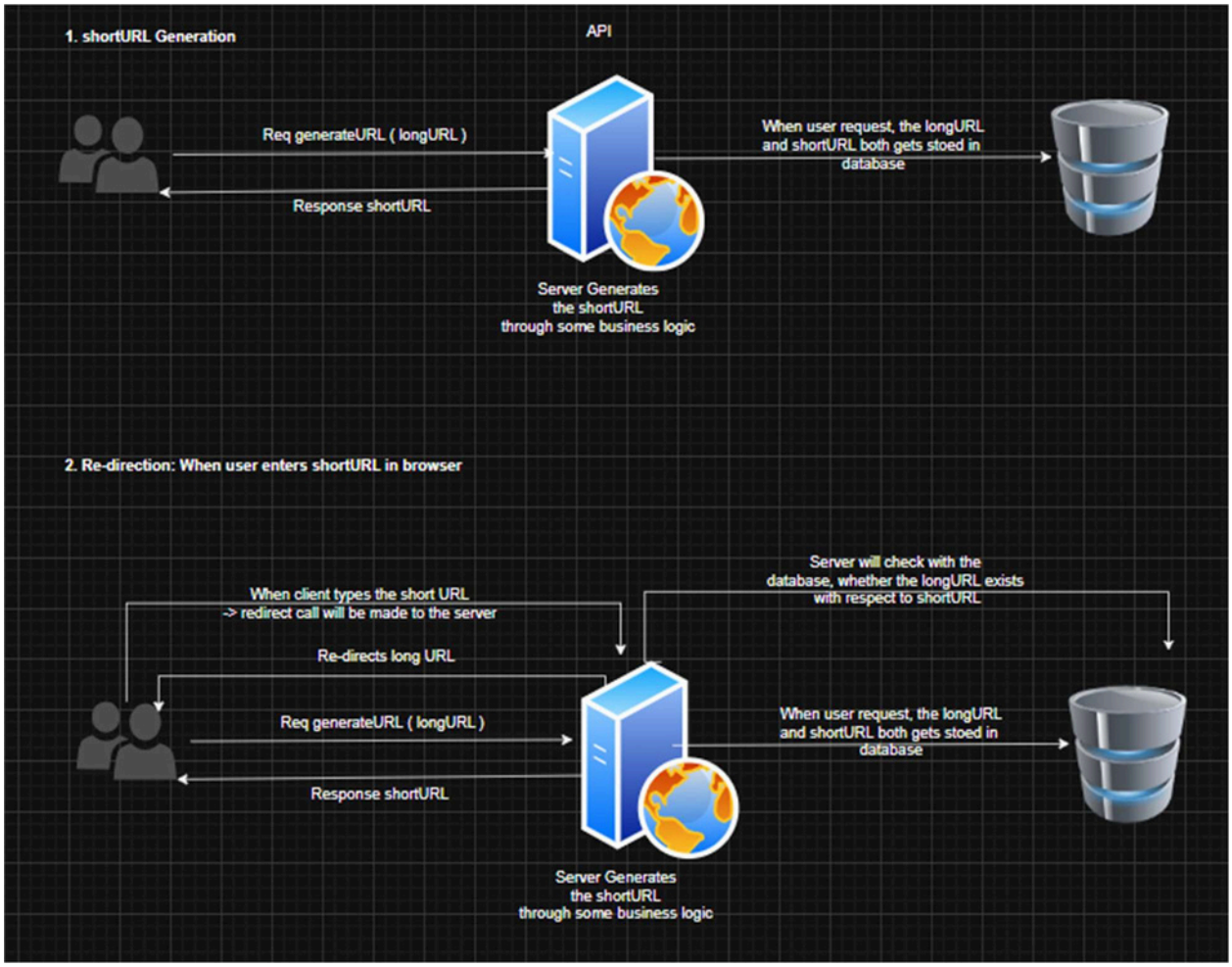
Flow:
Client → Server → Database → Server → Client

1. Workflow

1. Short URL Generation 1. Client sends long URL request.
2. Server generates short URL.
3. Server stores long URL + short URL in database.
4. Server responds with short URL.

2. Redirection

1. Client enters short URL.
2. Server validates short URL from database.
3. Server redirects to original long URL.



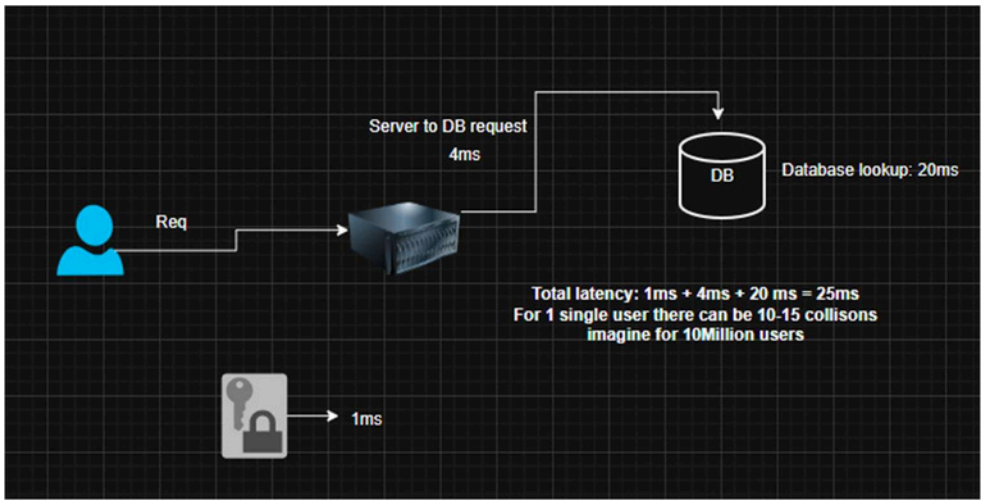
5.Low Level Design (LLD):-

Approach 01: Hashing (Encryption)

Method: shortURL = encrypt(longURL)
Algorithms: MD5, SHA-1, Base64

Problems:

1. Long hash length (notshort).
2. High collision probability.
3. Multiple DB lookups.
4. High latency at scale.
   Not suitable for large-scalesystems



<u>Approach 02:</u> Counter-Based Approach (Recommended)

Logic:

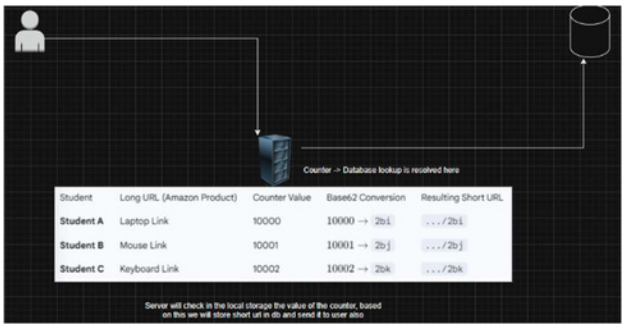
1. Generate a global counter value.
2. Convert counter to Base62.
3. Use Base62 value as short URL.

Example:

Counter = 10000

Base62 = 2bi

Short URL = short.ly/2bi



Advantages

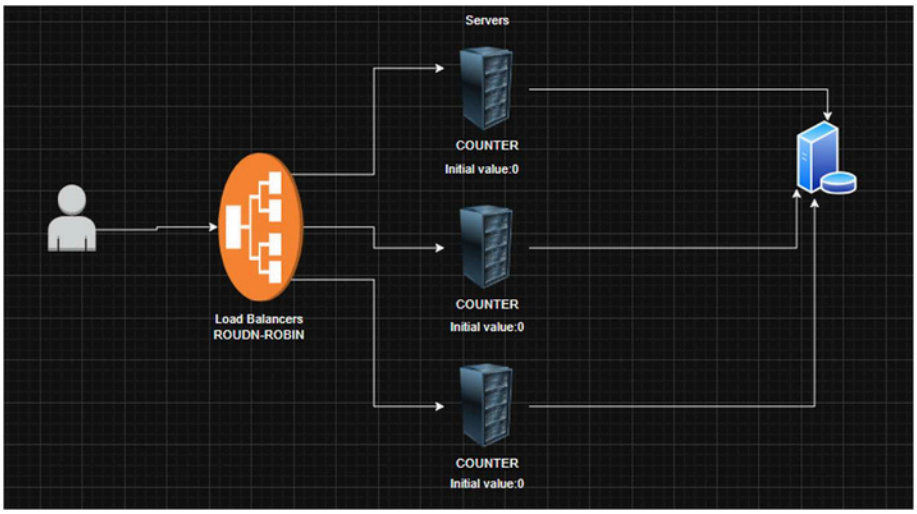
- No collision
- Fast generation
- Predictable length

Scaling Challenges & Solutions

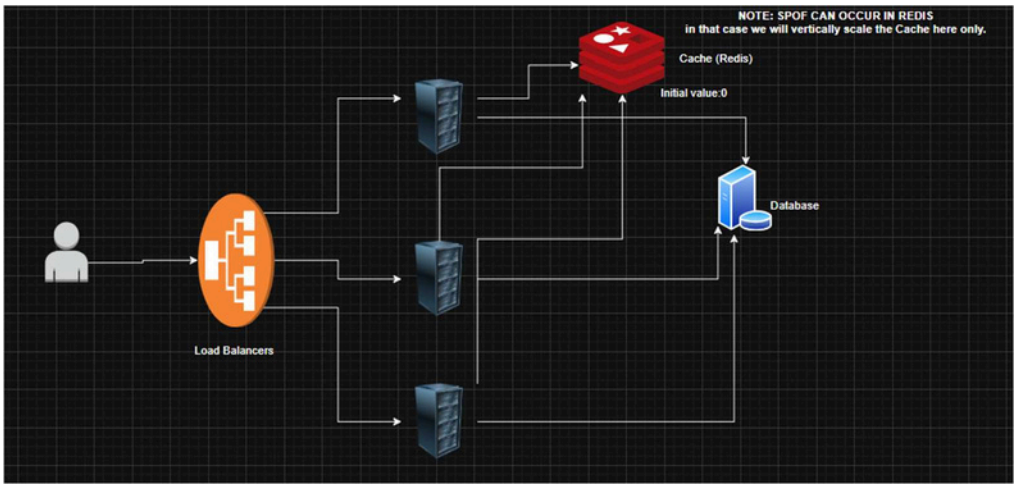
Problem1:Monolithic Server

- Cannot handle 100M users

Solution: Horizontal Scaling (Multiple Servers)



Problem 2: Distributed Counter Conflict

- Multiple servers may generate same counter.

Solution:
Use Global Counter in Cache (Redis) with atomic increment.